

E-Commerce Sales Analysis Using SQL

1. Project Overview

This project focuses on analyzing an E-Commerce database using sql to derive meaningful business insights which could be helpful in business administration and expansion. The objective is to evaluate the sales performance throughout the year, customer division, category wise trends and payment method usages to support data-driven decision making.

Tools used : MySQL,SQL(Joins,Aggregations,Subqueries)

2. Database Schema

The database is divided into five normalised tables :customers, products, categories, orders and order_items. The schema is based on real world design of a E-commerce platform where customers place order and each order contains products which are divided into respective categories

3. Project Overview

- Which product categories generate the highest revenue?
- How do orders and revenue vary across the countries?
- What are the monthly trends in orders and revenue?
- Which payment methods are most commonly used?
- What is the average order value?

4. Analysis Summary

Category wise analysis revealed that Home & Kitchen category contribute the most number of orders whereas the clothing sector generates the most revenue.

Country wise analysis showed that most orders come from Germany followed by the UK and also highest revenue is generated from these countries respectively.

Monthly trend analysis highlighted the seasonal patterns in customer purchases and how the sales are highest during the festive seasons, while payment method analysis showed a strong preference for cashless payment options

5. Key Findings

- Digital payments dominate both order volume and revenue.
- Sales show clear seasonal patterns with peak performance toward year-end
- Revenue concentration exists in specific categories and countries.
- Average Order Value remains consistent across segments, indicating stable pricing.
- High-revenue regions and categories should be prioritized for targeted marketing.

6. Conclusions

This project used SQL to analyze an e-commerce database and extract key business insights related to sales, customers, and revenue trends. The analysis showed a strong preference for digital payment methods, high revenue contribution from categories like Clothing and Books, and significant regional differences in customer spending. Monthly trends revealed seasonal sales patterns with higher performance toward the end of the year. Overall, the project demonstrates the ability to convert raw data into actionable insights using SQL.